

Homework 2:

An isothermal reversible reaction ($A \leftrightarrow B$) is carried out in a liquid solution. The reaction is first-order in both directions. The forward rate constant is k_1 (h^{-1}), the backward rate constant is k_{-1} (h^{-1}), the equilibrium constant is 1. The inlet molar flow rate to the plant is F_{A0} (mol/m^3) and enters at rate v_0 (m^3/h). The reactor is a stirred tank of volume V m^3 . The exit stream then goes to a separator, where one stream of v_R (m^3/h) with n times ($n > 1$) the concentration of A at the exit of the reactor (C_A) is sent back to the CSTR in a continuous operation. The rest of the stream from the separator becomes the system exit (no reaction occurs in the separator).

- Find the system conversion with respect to k_1 , v_0 , v_R and V, n .
- Compare this conversion with what you will get if there is no recycle.